NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) Date:31-12-24
Time: 3HRS GTM-7 Max. Marks: 300

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
A	D	B	D	A	A	B	A	C	B
11	12	13	14	15	16	17	18	19	20
C	A	B	C	B	B	C	D	A	B
21	22	23	24	25					
5	5	200	8	1					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
D	B	B	C	B	C	B	A	A	B
36	37	38	39	40	41	42	43	44	45
A	D	C	C	D	B	A	B	B	A
46	47	48	49	50					
4	4	890	12	8					

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MATHEMATICS

51	52	53	54	55	56	57	58	59	60
B	C	D	C	D	B	B	A	D	C
61	62	63	64	65	66	67	68	69	70
B	D	A	C	B	B	C	C	A	D
71	72	73	74	75					
60	3	6	36	5					

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PHYSICS

1. $\frac{1}{f}=\left(\frac{1.5}{1.75}-1\right)\left(-\frac{2}{R}\right)\Rightarrow f=3.5R$

2. $\frac{1}{2}mv^2=\frac{hc}{\lambda}-W$

When the wavelength is changed to $\frac{3\lambda}{4}$,

$\frac{1}{2}mv'^2=\frac{hc}{\left(\frac{3\lambda}{4}\right)}-W=\frac{4}{3}\left(\frac{hc}{\lambda}-W\right)+\frac{W}{3}$

$\Rightarrow\frac{1}{2}mv'^2=\frac{4}{3}\left(\frac{1}{2}mv^2\right)+\frac{W}{3}$

$\Rightarrow v'^2>\frac{4}{3}v^2$

$\Rightarrow v'>\left(\frac{4}{3}\right)^{1/2}v$

3. For second excited state, n=3

$L=\frac{nh}{2\pi}\Rightarrow L\propto n\Rightarrow L_H=L_{Li}$

$E\propto\frac{Z^2}{n^2}\Rightarrow|E_{Li}|>|E_H|$

The magnitude of energy of lithium ion is more than the magnitude of energy of hydrogen atom

4. Resistance of galvanometer, $G=\frac{v}{i_g}=\frac{75\times10^{-3}}{15\times10^{-3}}=5\Omega$

$i_g=\frac{150mA}{10}=15mA$

$=15\times10^{-3}A$

$total\ current\ i=6\times150=900A$

$(i-i_g)S=i_gG$

$S=\frac{i_gG}{i-i_g}=\frac{(15\times10^{-3})5}{150\times6-15\times10^{-3}}=8.3\times10^{-5}\Omega$

$=0.083\times10^{-3}=8.3\times10^{-5}$

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5. Torque about Q of charge – q is zero, so angular momentum charge – q is constant, but distance between charges is changing, so force is changing, so speed and velocity are changing

6. Conceptual

7. $B_D = B_{AC} - B_{CB} = \frac{\mu_0}{4\pi}\left[\frac{i}{R}\sin 60°\right] - \frac{\mu_0}{4\pi}\left[\frac{i}{\sqrt{3}R}\sin 30°\right]$

$= \frac{\mu_0 i}{4\pi R\sqrt{3}}$

8 . Linear speed of particle at t =2s is

$v = (2)^2 = 4 m/s$

Radial acceleration, $a_r = \frac{v^2}{r} = \frac{4^2}{2} = 8\ m/s$

The tangential acceleration is $a_t = \frac{dv}{dt} = 2t$

Tangential acceleration at t =2 s is $a_t = (2)(2) = 4 m/s^2$

Net acceleration of particle at t = 2s is

$a = \sqrt{(a_r)^2 + (a_t)^2}$

$= \sqrt{(8)^2 + (4)^2}$

$a = \sqrt{80}\ m/s^2$

9. At $t = 5s, q = 9, i = \frac{dqV}{dt} = 2, V_a - \frac{q}{C} - L\frac{di}{dt} - iR = V_b \Rightarrow V_a - V_b = 46.5V$

10. $\rho = R \times length$ and $\varepsilon_0 = \frac{C}{length} \Rightarrow \frac{CV}{\rho\varepsilon_0} = \frac{V}{R} = I$ (current)

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11. $Y = \overline{\overline{(A.\overline{(A.B)}).((B.\overline{(A.B)})}} = A.\overline{(A.B)} + B.\overline{(A.B)}$

$= A.\left(\overline{A} + \overline{B}\right) + B.\left(\overline{A} + \overline{B}\right)$

$= A.\overline{A} + A.\overline{B} + \overline{A}.B + B.\overline{B}$

$= A.\overline{B} + \overline{A}.B$

When A=1,B = 0 then Y =1. $\overline{0}$+ $\overline{1}$.0=1

When A=1, B =1 then Y=1. $\overline{1}$+$\overline{1}$.1=0

When A=0,B = 0 then Y =0. $\overline{0}$+$\overline{0}$.0=0

12. $P = Fv$ = constant

$\therefore F = \frac{P}{v}$

$\therefore mv\left(\frac{dv}{ds}\right) = \frac{P}{v}$

$\therefore \int_0^s ds = \frac{m}{P}\int_v^{2v} v^2 dv$

Solving we get, $s = \frac{7mv^3}{3P}$

13. Conceptual

14. Let ΔT be the increase in temperature when the volume of the gas is changed by ΔV at constant pressure. The change in internal energy of n moles of a gas is given by

$\Delta U = nC_v\Delta T$ .....(i)

We know that $C_p - C_v = R$

Or $\frac{C_p}{C_v} = 1 + \frac{R}{C_v}$. $But \frac{C_p}{C_v} = \gamma$

Therefore, $\gamma = 1 + \frac{R}{C_v}$, which gives

$C_v = \frac{R}{(\gamma - 1)}$ ..... (ii)

Also PV = n RT. At constant pressure, when volume changes by AV, the change in temperature ΔT is given by

$\Delta T = p\frac{\Delta V}{nR}$

(Δ V=2V-V=V) (iii)

Using (ii) and (iii) in (i) we have

$\Delta U = n \times \frac{R}{(\gamma - 1)} \times \frac{PV}{nR} = \frac{PV}{(\gamma - 1)}$

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15. $\frac{1}{v}-\frac{1}{u}=\frac{1}{f} \Rightarrow \frac{-1}{1+f}+\frac{1}{f-1}=\frac{1}{f} \Rightarrow f=1+\sqrt{2}$

16. $T=2(1+2+1)s$.

$=8\text{s}$

17. Maximum frictional force on block B is $\mu m_B g=0.4\times3\times10=12N$

Hence, maximum acceleration $=\frac{12}{3}=4\ ms^{-1}$

Hence, maximum force, $F=(m_A+m_B)a=(6+3)\times4=36N$

18. Conceptual

19. Four resistances carrying a current shown in Figure are immersed in a box containing ice at 0°C.

Net resistance of circuit, $R=\frac{10\times5}{10+5}+\frac{10\times5}{10+5}=\frac{20}{3}\Omega$

Heat generated in circuit per minute is

$Q=1^2Rt=(10)^2\text{ x }\frac{20}{3}(10\text{ x }60)=4\text{ x }10^5\text{ J}=\frac{4\times10^5}{4.2\times10^3}\ k\ cal$

$m=\frac{Q}{L}=\frac{4\times10^5}{4.2\times10^3\times80}=1.19\ kg$

20. $r\propto A^{\frac{1}{3}} \Rightarrow \frac{r_1}{r_2}=\left(\frac{A_1}{A_2}\right)^{1/3}$

$\Rightarrow \frac{3}{5}=\left(\frac{27}{A}\right)^{1/3} \Rightarrow \frac{27}{125}=\frac{27}{A} \Rightarrow A=125$

Number of nuclei in atom X

$=A-52=125-52=73$

21. $B=\mu_0 ni$

$B=\frac{\mu_0\times500\times3}{0.4}$

Torque acting on the second coil

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$\tau = M\ B\sin 90^o$

$= (N_2\ i_2\ A_2)B = N_2\ i_2 \times A_2 \times \mu_0 n \times i$

$\tau = \frac{3}{5} \times 10^{-7}\ Nm;\ x = 5$

22. $\frac{translational\ KE}{total\ KE} = \frac{\frac{1}{2}mv^2}{\frac{1}{2}Iw^2 + \frac{1}{2}mv^2}$

$= \frac{\frac{1}{2}}{\frac{1}{2} + \frac{1}{5}} = \frac{\frac{1}{2}}{\frac{5+2}{10}} = \frac{1}{2} \times \frac{10}{7}$

23. In equilibrium, the pressures at the two surfaces should be equal as they lie in the same horizontal level. If the atmospheric pressure is P_0 and a force F is applied to maintain the equilibrium, the pressure just below the pistons are

$P_M = P_0 + \frac{10N}{1\ cm^2}\ and\ P_N = P_0 + \frac{F}{20\ cm^2}$

$P_M = P_N$

$= \frac{10}{1} = \frac{F}{20} = 200N$

These pressures should be the same, i.e., $P_M = P_N$ which gives F = 200 N.

24. $E_y = \frac{k\lambda}{r}(sin\ 60° + sin\ 30°) = \frac{k\lambda}{r}\left(\frac{\sqrt{3}}{2} + \frac{1}{2}\right) = \frac{k\lambda(\sqrt{3}+1)}{2r}$

$E_x = \frac{k\lambda}{r}(cos\ 60° - cos\ 30°) = \frac{k\lambda}{r}\left(\frac{1}{2} - \frac{\sqrt{3}}{2}\right) = \frac{k\lambda(1-\sqrt{3})}{2r}$

$E = \sqrt{E_x^{\ 2} + E_y^{\ 2}}$

$= \frac{k\lambda}{2r}\sqrt{(\sqrt{3}+1)^2 + (1-\sqrt{3})^2}$

$E = \frac{k\lambda}{2r}\sqrt{8}$

25. Resistance of each part = R/2n

For 'n' such part connected in series, equivalent resistances, say

$R_1 = n = \left[\frac{R}{2n}\right] = \frac{R}{2}$

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Similarly, equivalent resistance, Say R_2, for another set of n identical, respectively, in Parallel resistances would be

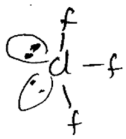
$$\frac{1}{n}\left(\frac{R}{2n}\right)=\frac{R}{2n^2}$$

For getting maximum of R_1 and R_2, the resistances should be connected in series and hence,

$$R_{eq} = R_1 + R_2 = \frac{R}{2}\left(1+\frac{1}{n^2}\right)$$

CHEMISTRY

26.



$$\frac{7-3}{2}=\frac{4}{2}=2$$

$$IF_5 \rightarrow \frac{7-5}{2}=\frac{2}{2}=1$$

$$ClF \rightarrow \frac{7-1}{2}=3$$

CCl_4 zero lone pair on central atom

27. H_2O has more dipole moment i.e. 1.85D than H_2S $(\mu = 0.95D)$

28. $CH_3-\underset{\underset{\underset{O}{\|}}{C-NH_2}}{CH}-CH_2-CH_2-OH$

- $CONH_2$ amide group functional group
- OH – sec – prefix.

29. More substituted Alkene will be the major product (saytzeff's rule)

30. $w=\frac{E\times C\times t}{96500}$; t= time, E=Eq.wt; C=Current in amp

$$w=\frac{31.75\times 20\times 9650}{96500}=31.75\times 2=63.5g$$

31. $\underset{1\text{ mole}}{C_2H_6}+\frac{7}{2}O_2 \rightarrow \underset{2\text{ mole}}{2CO_2}+3H_2O$

30g---------- $2\times 22.7lit$

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? …….. $2.27 lit$

$\frac{2.27 \times 30}{2 \times 22.7} = 1.5 g$

32. $Ag^{+} / Ag \; E^{o} = 0.8$

Ag^{+} / Ag S.R.P. is more than Mg^{+2} / Mg SRP

33.

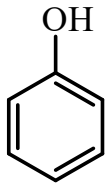
34. Neon radius is greater than fluorine

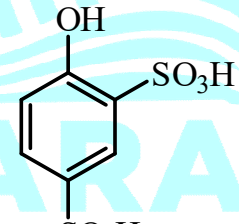
35. It is a nucleophilic substitution reaction 1-phenyl-1-propanol will be most reactive because of formation of more stable benzyl type of carbocation.

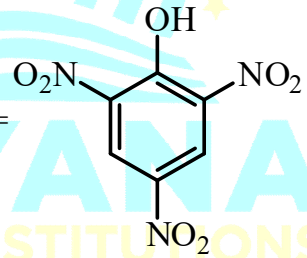
$$C_6H_5 - \underset{\underset{OH}{|}}{CH} - CH_2 - CH_3 \rightarrow C_6H_5 - \overset{+}{C}H - CH_2 - CH_3$$

36. In a octahedral crystal field d-electrons enters in to less energetic t_2g set before eg set of orbitals.

37.

A= 

B= 

C= 

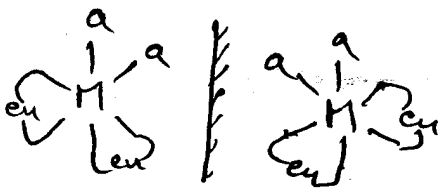
38. B has highest M.P. born not reacts with acids to liberate H_2 gas.

$\rightarrow In^{+3}$ can exists but B^{+3} can not exists.

$\rightarrow$ En B-2; Al-1.5, Ga-1.6

In – 1.7; Tl – 1.8

39. $cisMa_2(AA)_2$ can exhibit optical isomerism

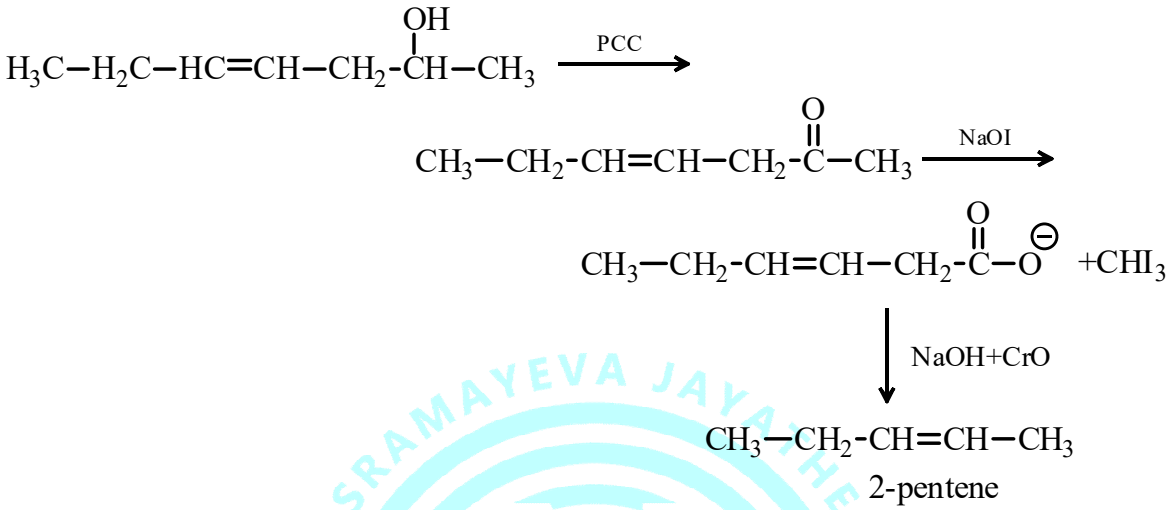


Optical isomers (Enantiomers)

40. Arginine, isolcucine, Leucine, Valine, Lysine Threonine. Methionine Phenylalanine, Tryptophan and Histidine are essential amino acids.

41.

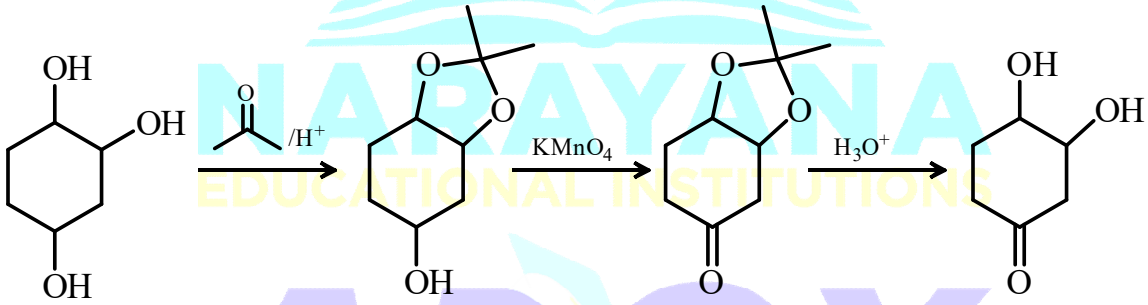
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42. Cu^{+2} Reduction potential is more it can not reduce H^+ ions other ions are strong reducing agents then H^+.

43. Conceptual

44.

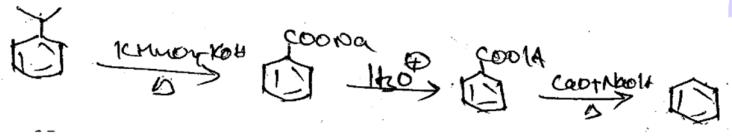


45. Due to high lattice enthalpy $BaSO_4$ insoluble in water.

46. $n = 4, l = 1$ Means 4p

$m_l = -1 \& +1$

$|m| = -1 \& +1$ 2 orbitals so $4e^-$ can present.

47. 

In benzene no of σ bonds -12 no. of π -bonds – 3

Ratio $\frac{12}{3} = 4$

48. $CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)}$

$\Delta_r H^o = \Delta_f H^o_{(\text{products})} - \Delta_f H^o_{(\text{reactants})}$

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$$=\left[\Delta_f H^o_{(co_2)}+2\Delta H^o_f\left(H_2O\right)\right]-\Delta_f H^o_{(CH_4)}+2\Delta_f H^o_{(O_2)}$$
$$=\left(-393.5+2\times-285.8\right)-\left(-75+2(o)\right)$$
$$=-393.5-571.6+75$$
$$=890.1\approx 890kJ\,/\,mole$$

49. (i) Rate $=k[A][B]^2$

When $[A]=0.1$ mol/lit, $[B]=0.2$ mol/lit

Rate $=2.0\times10^{-6}\times(0.1)(0.2)^2$

$=8\times10^{-9}\ \therefore x=8$

(ii) When $[A]=0.06$ mol/lit

$\therefore$A reacted = 0.1-0.06=0.04

B reacted = 0.02

$[B]=0.2-0.02=0.18$ mol/lit

Rate $=2.0\times10^{-6}(0.06)(0.18)^2$

$=3.888\times10^{-9}\approx4\times10^{-9}\ \ mol\,L^{-1}s^{-1}$

$x=8, y=4$

$x+y=8+4=12$

50. $\dfrac{\Delta P_1}{\Delta P_2}=\dfrac{\chi_{(1)(\text{solute})}}{\chi_{2(\text{solute})}}$; $\chi_{2(\text{solute})}=\dfrac{\Delta P_2\times\chi_{1(\text{solute})}}{\Delta P_1}$

$=\dfrac{20\times0.1}{10}=0.2$

$\chi_{solvent}=1-\chi_{2(\text{solute})}=1-0.2=0.8$

So, 8×10^{-1};

$a=8$

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MATHEMATICS

51.

$$f_n(x)=\begin{cases}|n-2|+\sqrt{x+2} & ; \quad -2<x\le-1\\ |n-1|+\sqrt{x+1} & ; \quad -1\le x<0\\ |n|+\sqrt{x} & ; \quad 0\le x<1\end{cases}$$

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At

$x=-1$

$LHL=|n-2|+1$ $\qquad |n-2|+1=|n-1|$

$RHL=|n-1|$ $\qquad \Rightarrow n\geq 2$

At

$x=0$

$LHL=|n-1|+1$

$RHL=|n|$

$|n-1|+1=|n|\Rightarrow n\geq 1$

52. (3)

By the given data circle is $x^2+(y-k)^2=k^2$

$\Rightarrow\ x^2+y^2-2ky=0$ $\qquad$ ….. (1)

Differentiate 'x'

$$2x+2y\frac{dy}{dx}-2k\frac{dy}{dx}=0$$

$$\therefore\ x+y\frac{dy}{dx}=k\frac{dy}{dx}$$

$$\therefore \text{ Eqs. (1) } x^2+y^2-2\left(\frac{x+y\frac{dy}{dx}}{\frac{dy}{dx}}\right)y=0.$$

$$(x^2+y^2)\frac{dy}{dx}-2xy-2y^2\frac{dy}{dx}=0$$

$$\Rightarrow\ (x^2-y^2)\frac{dy}{dx}-2xy=0$$

$$\therefore\ (x^2-y^2)dy-2xy\,dx=0$$

53. (4)

$$\log_{\sqrt{3}/2}\frac{|z-2|+5}{4|z-2|-4}<\log_{\sqrt{3}/2}\left(\frac{\sqrt{3}}{2}\right)^2$$

$$\therefore\ \frac{|z-2|+5}{4(|z-2|-1)}>\frac{3}{4}\ \text{ and }\ 4(|z-2|-1)>0$$

$$\therefore\ |z-2|<4\ \text{ and }\ |z-2|>1$$

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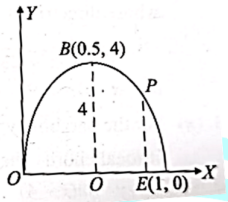
$\therefore\ 1<|z-2|<4$

Clearly area is enclosed between two concentric circles with centre (2, 0) and radius 1 and 4

Required area = $\pi(16)-\pi(1)=15\pi$

54. (3)

7. (c) The path of the water jet is a parabola.
Let its equation be
$y = ax^2 + bx + c$
It should pass through (0, 0), (0.5, 4), and (1, 0). Therefore,



$c = 0,\ a = 16,\ b = 16$

$\therefore\quad y = -16x^2 + 16x$

If $x = 0.75$, then $y = 3$.

Length of latus rectum = 4(distance from S(4, 5) to Eqs. (3)) = $2\sqrt{2}$.

55. (4)

By verification

56. $d_{n+1} = 2d_n + n.(1 + 2^n)$

$$\Rightarrow \frac{d_{n+1}}{2^{n+1}} = \frac{d_n}{2^n} + \frac{n}{2^{n+1}} + \frac{n}{2}$$

$$\Rightarrow \frac{d_2}{2^2} = \frac{d_1}{2} + \frac{1}{2^2} + \frac{1}{2}$$

$$\Rightarrow \frac{d_3}{2^3} = \frac{d_2}{2^2} + \frac{2}{2^3} + \frac{2}{2}$$

$$\Rightarrow \frac{d_4}{2^4} = \frac{d_3}{2^3} + \frac{3}{2^4} + \frac{3}{2}$$

$$\Rightarrow \frac{d_n}{2^n} = \frac{d_{n-1}}{2^{n-1}} + \frac{n-1}{2^n} + \frac{(n-1)}{2}$$

Adding

$$\Rightarrow \frac{d_n}{2^n} = \frac{d_1}{2} + \left(\frac{1}{2^2} + \frac{2}{2^3} + \frac{3}{2^4} + ... + \frac{n-1}{2^n}\right) + \frac{1}{2}\left[1 + 2 + ... + (n-1)\right]$$

$d_n = (n^2 - n + 6)\ 2^{n-2} - n - 1$

$\Rightarrow$ given $d_k = (k^2 - 2k + 13)\ 2^{k-2} - k - 1 \Rightarrow k = 7$

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57. (2)

$\vec{r}\cdot(\vec{a}+\vec{b}+\vec{c})=0 \Rightarrow [\vec{a}\ \vec{b}\ \vec{c}](\sin x+\cos y+2)=0$

$\therefore \sin x+\cos y+2=0$ which is possible only when $\sin x=-1$ and $\cos y=-1$

For (x^2+y^2) to be minimum $x^2=\frac{\pi^2}{4}$ and $y^2=\pi^2$

$\therefore \frac{20}{\pi^2}(x^2+y^2)_{\min}=\frac{20}{\pi^2}\left(\frac{\pi^2}{4}+\pi^2\right)=25.$

58. (1)

$(a^2+ab+1)(1)>0 \quad \forall a\in R$

$\therefore \Delta<0 \Rightarrow b^2-4<0 \Rightarrow b\in(-2,2)$

59. (4)

(h,k) is the foot of perpendicular from (12, 10) on to the line $5x+y-18=0$

$\therefore \frac{h-12}{5}=\frac{k-10}{1}=-\left(\frac{10+60-18}{26}\right)$

$\therefore (h,k)=(2,8).$

60. (3)

Required term $={}^{10}C_5(x\sin\alpha)^5\left(\frac{\cos\alpha}{x}\right)^5$

$\therefore {}^{10}C_5\left(\frac{\sin 2\alpha}{2}\right)^5\le {}^{10}C_5\frac{1}{2^5}$

61.

(d) $f(x)=\frac{\sqrt{x^2-25}}{4-x^2}+\log_{10}(x^2+2x-15)$

for domain: $x^2-25\ge 0\Rightarrow x\in(-\infty,-5)\cup(5,\infty)$

$4-x^2\ne 0 \Rightarrow x\ne\{-2,2\}$

and $x^2+2x-15>0\Rightarrow(x+5)(x-3)>0$

$\Rightarrow x\in(-\infty,-5)\cup(3,\infty)$

$\therefore x\in(-\infty,-5)\cup[5,\infty)$ is the domain

Hence $\alpha=-5;\beta=5\Rightarrow\alpha^2+\beta^3=150$

62. Any point on the parabola $y^2=4ax$ is $\left(at^2,2at\right)$. Equation of chord of the ellipse

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$\frac{x^2}{2a^2}+\frac{y^2}{a^2}=1$, whose mid-point is $\left(at^2,2at\right)$

$\frac{x.at^2}{2a^2}+\frac{y.2at}{a^2}=\frac{a^2t^4}{2a^2}+\frac{4a^2t^2}{a^2}$

$\Rightarrow \quad tx+4y=at^3+8at \qquad \left(\because t\neq 0\right)$

As it passes through $\left(11a,-\frac{a^2}{4}\right)$,

$\Rightarrow \quad 11at-4\left(\frac{a^2}{4}\right)=at^3+8at$

$\Rightarrow \quad at^3-3at+a^2=0$

$\Rightarrow \quad t^3-3t+a=0\left(a\neq 0\right) \qquad \ldots(\text{i})$

Now, three chords of the ellipse will be bisected by the parabola if the equation (i) has three real and distinct roots.

Let $f\left(t\right)=t^3-3t+a$

$f'\left(t\right)=3t^2-3=0$

$\Rightarrow \quad t=\pm 1$

So, $f\left(1\right)f\left(-1\right)<0$

$\Rightarrow\ a\in\left(-2,2\right)$

But $a\neq 0$, so $a\in\left(-2,0\right)\cup\left(0,2\right)\Rightarrow a=1$ or -1

Also, $S_{11}<0$

$\frac{t^4}{2}+4t^2-1<0$

For $a=1$ or -1,

A root of $f\left(t\right)=0$ is >1

$\therefore$ Number of integral values of $a=0$.

63. (3)

$P=(-s-2,2s-6,1)$; $Q=(-3t-2,4t+2,2t+5)$

Distance $\overrightarrow{PQ}=(3t-s,2s-4t-8,-2t-4)$

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Again distance $\overrightarrow{PQ}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -1 & 2 & 0\\ -3 & 4 & 2\end{vmatrix}=4\hat{i}+2\hat{j}+2\hat{k}=(2,1,1)$

$\therefore \dfrac{3t-s}{2}=\dfrac{2s-4t-8}{1}=\dfrac{-2t-4}{1}$

$\therefore t=-1, s=1$

$\therefore P=(-3,-4,1)$

Equation $\overrightarrow{PQ}=\dfrac{x+3}{2}=\dfrac{y+4}{1}=\dfrac{z-1}{1}$

But $(3,h,k)$ lie on it

$3=h+4=k-1,\ h=-1, k=4$

$\therefore h+k=3$

64. (3)

By the given data $\begin{vmatrix}1 & k & 3\\ 3 & k & -2\\ 2 & 4 & -3\end{vmatrix}=0$

$\Rightarrow -3k+8-k(-9+4)+3(12-2k)=0 \Rightarrow k=11$

Put $z=t$

$\therefore$ Given equation $x+11y=-3t;\ 3x+11y=2t$ by solving $2x=5t$

$\Rightarrow x=\dfrac{5t}{2},\ y=\dfrac{-3z-x}{11}=\dfrac{-t}{2}$

$\therefore \dfrac{xz}{y^2}=\dfrac{\left(\frac{5t}{2}\right)t}{\frac{t^2}{4}}=10.$

65. (2)

$n(S)=4;$ $[(\alpha,\beta)=(1,1),(1,0),(0,0),(w,w^2)]$

$n(E)=2$ $[(\alpha,\beta)=(1,1),(0,0)]$

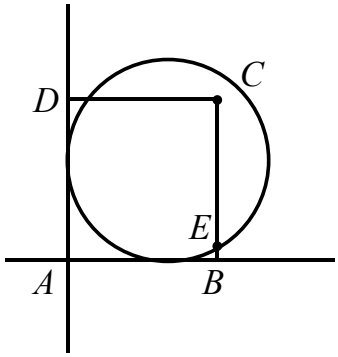
$P(E)=\dfrac{1}{2}$

66. (2)

Equation of circle is $x^2+y^2-2rx-2ry+r^2=0$

But E(32, 1) lie on it

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$\therefore\ r^2-66r+1025=0$

$\therefore\ r=25$ or 41

$r=25$ (reject $r=41$)

67. (3)

$$\int_0^{\alpha}\frac{t^{50}}{1-t}dt=\int_0^{\alpha}\frac{1-(1-t^{50})}{1-t}dt$$

$$=\int_0^{\alpha}\frac{1}{1-t}dt-\int_0^{\alpha}(1+t+t^2+.....+t^{49})dt$$

$$=-\left[\log_e|1-t|\right]_0^{\alpha}-\left(t+\frac{t^2}{2}+....+\frac{t^{50}}{50}\right)_0^{\alpha}$$

$$=-\log_e|1-\alpha|-P_{50}(\alpha)$$

$$=-\beta-P_{50}(\alpha).$$

68. (3)

$$A=\begin{pmatrix}0&-1&-2\\1&0&-1\\2&1&0\end{pmatrix},$$

$\therefore\ |A|=0$ ($\because$ A is skew symmetric matrix)

69. (1)

$x=\sec\theta-\cos\theta$

$x^2+4=(\sec\theta-\cos\theta)^2+4$

$x^2+4=(\sec\theta+\cos\theta)^2$ and $y^2+4=(\sec^5\theta-\cos^5\theta)^2+4=(\sec^5\theta+\cos^5\theta)^2$

$$\frac{dy}{dx}=\frac{dy/d\theta}{dx/d\theta}=\frac{5\sec^4\theta\cdot(\sec\theta\tan\theta)+5\cos^4\theta\sin\theta}{\sec\theta\tan\theta+\sin\theta}=\frac{5[\sec^5\theta+\cos^5\theta]}{\tan\theta(\sec\theta+\cos\theta)}$$

$$\left(\frac{dy}{dx}\right)^2=\frac{25\cdot(\sec^5\theta+\cos^5\theta)^2}{(\sec\theta+\cos\theta)^2}=\frac{25(y^2+4)}{x^2+4}$$

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$\therefore\ (x^2+4)(y')^2 = 5^2(y^2+4)$

$\therefore\ k = 5$

70. (4)

$$\frac{P}{Q^2} = \frac{\left(x-\frac{1}{x}\right)\left(x^2+\frac{1}{x^2}+1\right)}{\left(x-\frac{1}{x}\right)^2} = \frac{\left(x-\frac{1}{x}\right)^2+3}{\left(x-\frac{1}{x}\right)}$$

$$\frac{P}{Q^2} = \left(x-\frac{1}{x}\right)+\frac{3}{\left(x-\frac{1}{x}\right)}$$

$$\frac{P}{Q^2} \geq 2\sqrt{\left(x-\frac{1}{x}\right)\frac{3}{\left(x-\frac{1}{x}\right)}} = 2\sqrt{3}\ \text{(A.M. } \geq \text{ G.M)}$$

71. (60)

p(probability of getting white ball) $= \frac{40}{60} = \frac{2}{3} = p$

$q = \frac{1}{3}$ and $n = 30$

Mean= np $= 30\times\frac{2}{3} = 20$ and standard deviation= $\sqrt{npq} = \sqrt{30\times\frac{2}{3}\times\frac{1}{3}} = \sqrt{\frac{20}{3}}$

$$\therefore\ \frac{20}{\sqrt{20/3}} = \sqrt{60}$$

72.

Given differential equation $\frac{dy}{dx} - \frac{\tan 2x}{\cos^2 x}y = \cos^2 x$

I.F. $= e^{-\int\frac{\sin 2x}{\cos 2x\cos^2 x}dx} = e^{-\int\frac{2\sin 2x}{\cos 2x(1+\cos 2x)}dx} = e^{\int\frac{dt}{t(1+t)}} = e^{\int\left(\frac{1}{t}-\frac{1}{1+t}\right)dt}$

I.F. $= \frac{\cos 2x}{2\cos^2 x}$

Solutions is $y\left(\frac{\cos 2x}{2\cos^2 x}\right) = \int\frac{\cos 2x}{2\cos^2 x}\cdot\cos^2 x\,dx = \frac{\sin 2x}{4} + C$

at $x = \frac{\pi}{4}, y = \frac{3\sqrt{3}}{8} \Rightarrow C = 0$.

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$\therefore\ y=\frac{1}{2}\tan 2x\cos^2 x$

$\therefore\ k=\frac{1}{2}\Rightarrow 4k=2$

73.

By the given data

Let $f(x)=x^2+bx+c$ ..... (1)

$f(0)=0\ \Rightarrow\ c=0$

$f(1)=1\ \Rightarrow\ 1+b=1\ \Rightarrow\ b=0$

$\therefore\ \underset{x\to 0}{Lt}\frac{x^2(2^{\sin x}+3^{\tan x}-2)}{x-\sin x}=k\ln k$

$\Rightarrow\ \underset{x\to 0}{Lt}\frac{x^2(2^{\sin x}-1+3^{\tan x}-1)}{\left(x-\left(x-\frac{x^3}{3!}+....\right)\right)}=k\ln k$

$=\ \underset{x\to 0}{Lt}\frac{x^2}{\left(\frac{x^3}{3!}+.....\right)}\left(\frac{2^{\sin x}-1}{\sin x}\cdot\frac{\sin x}{x}\cdot x+\frac{3^{\tan x}-1}{\tan x}\cdot\frac{\tan x}{x}\cdot x\right)=k\ln k$

$=\ 6(\ln 2+\ln 3)=k\ln k$

$\therefore\ 6\ln 6=k\ln k$

$\therefore\ k=6$

74.

Let $y=\sqrt{-3+4x-x^2}$ = O.B

$y^2=-3+4x-x^2$

$\Rightarrow\ x^2+y^2-4x+3=0$ and the point is $(5,-4)$.

Clearly given value is the square of the distance between the circle

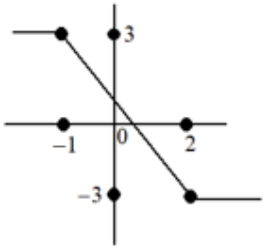
$x^2+y^2-4x+3=0$ and the point $(5,-4)$

Maximum = $(CP+r)^2=(5+1)^2=36$ $\quad[\because\ \text{CP}=5,\ r=1]$

75.

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Let $f(x)=|x-2|-|x+1|=\begin{cases}-x+2+x+1, & x<-1\\ -x+2-x-1, & -1\le x\le 2\\ -x-2-x-1 & x>2\end{cases}$



$=\begin{cases}3 & x<-1\\ -2x+1 & -1\le x\le 2\\ -3 & x>2\end{cases}$

$x\in\{-2,-1,0,1,2\}$ for exactly one solution

$\therefore$ Integral solution of $k=5$

SRAMAYEVA JAYATHE
NARAYANA
EDUCATIONAL INSTITUTIONS
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